

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 October 2019 Worked Solutions

Oct 2019 paper rebuilt with the worked solution after each question.

11

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 October 2019

Oct 2019 | 11 questions | 75 marks

11

questions

75

marks

Answers

worked solution
after each
question

Question 1

Radians

1.

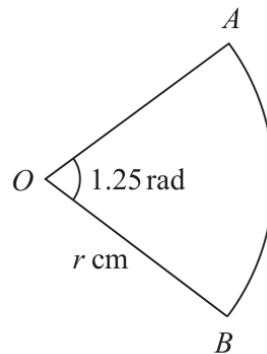


Figure 1

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is 1.25 radians.

Given that the area of the sector AOB is 15 cm^2

(a) find the exact value of r ,

(2)

(b) find the exact length of the perimeter of the sector. Write your answer in simplest form.

(3)

(Total 5 marks)

Worked Solution - Question 1

1. Use the sector area formula

For a sector, $A = \frac{1}{2}r^2\theta$. Here $A = 15$ and $\theta = 1.25 = \frac{5}{4}$, so $15 = \frac{1}{2}r^2 \cdot \frac{5}{4}$.

2. Find the exact radius

$r^2 = \frac{30}{5/4} = 24$, so $r = \sqrt{24} = 2\sqrt{6}$ because the radius is positive.

3. Find the perimeter

Arc length is $s = r\theta = (2\sqrt{6})\left(\frac{5}{4}\right) = \frac{5\sqrt{6}}{2}$. The perimeter is $2r + s = 4\sqrt{6} + \frac{5\sqrt{6}}{2} = \frac{13\sqrt{6}}{2}$.

Final answer

(a) $r = 2\sqrt{6}$ cm.

(b) $\frac{13\sqrt{6}}{2}$ cm.

Question 2

Straight Line

2. A tree was planted in the ground.

Exactly 2 years after it was planted, the height of the tree was 1.85 m.

Exactly 7 years after it was planted, the height of the tree was 3.45 m.

Given that the height, H metres, of the tree, t years after it was planted in the ground, can be modelled by the equation

$$H = at + b$$

where a and b are constants,

- (a) find the value of a and the value of b .

(4)

- (b) State, according to the model, the height of the tree when it was planted.

(1)

(Total 5 marks)

Worked Solution - Question 2

1. Form two equations

Using $H = at + b$: when $t = 2$, $2a + b = 1.85$; when $t = 7$, $7a + b = 3.45$.

2. Solve for the gradient of the model

Subtract the first equation from the second: $5a = 1.60$, so $a = 0.32$.

3. Find the starting height

Substitute into $2a + b = 1.85$: $0.64 + b = 1.85$, hence $b = 1.21$. At planting, $t = 0$, so $H = b = 1.21$ m.

Final answer

$a = 0.32$, $b = 1.21$. Height when planted: 1.21 m.

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10 marks

Question 3

Graphs of Functions

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

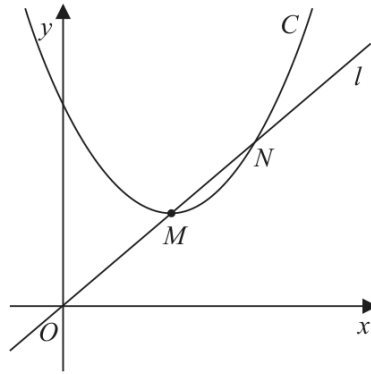


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = x^2 - 5x + 13$

The point M is the minimum point of C .

The straight line l passes through the origin O and intersects C at the points M and N as shown.

Find, showing your working,

(a) the coordinates of M , (3)

(b) the coordinates of N . (5)

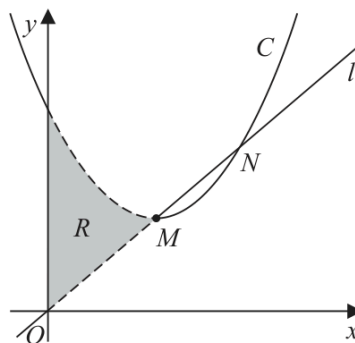


Figure 3

Figure 3 shows the curve C and the line l . The finite region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

(Total 10 marks)

Worked Solution - Question 3

Topic group

1. Complete the square to locate the minimum

$$y = x^2 - 5x + 13 = \left(x - \frac{5}{2}\right)^2 + \frac{27}{4}. \text{ Therefore } M = \left(\frac{5}{2}, \frac{27}{4}\right).$$

2. Find the equation of the line through O and M

The gradient of OM is $\frac{27/4}{5/2} = \frac{27}{10}$, so l has equation $y = \frac{27}{10}x$.

3. Find the second intersection

Solve $x^2 - 5x + 13 = \frac{27}{10}x$. This gives $x^2 - \frac{77}{10}x + 13 = 0$. One root is $x = \frac{5}{2}$, so the other is $x = \frac{26}{5}$. Then $y = \frac{27}{10} \cdot \frac{26}{5} = \frac{351}{25}$.

4. Describe the shaded region

The region is between the y-axis, the line, and the curve up to M : $0 \leq x \leq \frac{5}{2}$ and $\frac{27}{10}x \leq y \leq x^2 - 5x + 13$.

Final answer

$$M = \left(\frac{5}{2}, \frac{27}{4}\right), N = \left(\frac{26}{5}, \frac{351}{25}\right), \text{ and } 0 \leq x \leq \frac{5}{2},$$

$$\frac{27}{10}x \leq y \leq x^2 - 5x + 13.$$

Question 4

Basic Trigonometry

4. A parallelogram $ABCD$ has area 40 cm^2

Given that AB has length 10 cm , BC has length 6 cm and angle DAB is obtuse, find

- (a) the size of angle DAB , in degrees, to 2 decimal places, (3)
- (b) the length of diagonal BD , in cm , to one decimal place. (2)

(Total 5 marks)

Worked Solution - Question 4

1. Use the area of a parallelogram

Area = $ab \sin \theta$, so $40 = 10 \cdot 6 \sin \theta$. Hence $\sin \theta = \frac{2}{3}$.

2. Choose the obtuse angle

$\theta = 180^\circ - \sin^{-1}\left(\frac{2}{3}\right) = 138.19^\circ$ to 2 decimal places.

3. Use the cosine rule for BD

$BD^2 = 10^2 + 6^2 - 2(10)(6) \cos(138.19^\circ)$. Therefore $BD \approx 15.014\dots$, so $BD = 15.0$ cm to 1 decimal place.

Final answer

(a) $\angle DAB = 138.19^\circ$.

(b) $BD = 15.0$ cm.

Question 5

Differentiation

5. A curve has equation

$$y = \frac{x^3}{6} + 4\sqrt{x} - 15 \quad x \geq 0$$

- (a) Find $\frac{dy}{dx}$, giving the answer in simplest form.

(3)

The point $P\left(4, \frac{11}{3}\right)$ lies on the curve.

- (b) Find the equation of the normal to the curve at P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)

(Total 7 marks)

Worked Solution - Question 5

1. Rewrite before differentiating

$$y = \frac{x^3}{6} + 4x^{1/2} - 15.$$

2. Differentiate term by term

$$\frac{dy}{dx} = \frac{3x^2}{6} + 4 \cdot \frac{1}{2}x^{-1/2} = \frac{x^2}{2} + \frac{2}{\sqrt{x}}.$$

3. Find the tangent and normal gradients at P

At $x = 4$, $\frac{dy}{dx} = \frac{16}{2} + \frac{2}{2} = 9$. The normal gradient is therefore $-\frac{1}{9}$.

4. Build the normal equation

Using $P(4, \frac{11}{3})$: $y - \frac{11}{3} = -\frac{1}{9}(x - 4)$. Multiplying out gives $x + 9y - 37 = 0$.

Final answer

$$\frac{dy}{dx} = \frac{x^2}{2} + \frac{2}{\sqrt{x}}, \text{ and the normal is } x + 9y - 37 = 0.$$

Question 6

Graphs of Functions

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.

- (a) Sketch a graph of C , stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x -axis.

(3)

The line with equation $y = 10 - 2x$ is a tangent to C .

- (b) Find the possible values for k .

(5)

(Total 8 marks)

Worked Solution - Question 6

1. Read the graph features

For $y = \frac{4}{x} + k$, the horizontal asymptote is $y = k$. Setting $y = 0$ gives $0 = \frac{4}{x} + k$, so the x-intercept is $x = -\frac{4}{k}$.

2. Set the tangent line equal to the curve

A tangent touches once, so solve $10 - 2x = \frac{4}{x} + k$. Multiplying by x gives $2x^2 + (k - 10)x + 4 = 0$.

3. Use the discriminant condition

For one repeated intersection, discriminant = 0: $(k - 10)^2 - 4(2)(4) = 0$. Hence $(k - 10)^2 = 32$.

4. Find k

$k - 10 = \pm 4\sqrt{2}$, so $k = 10 \pm 4\sqrt{2}$. Both are positive, so both satisfy the condition in the question.

Final answer

Asymptote $y = k$, x-intercept $(-\frac{4}{k}, 0)$, and $k = 10 \pm 4\sqrt{2}$.

Question 7

Differentiation

7.

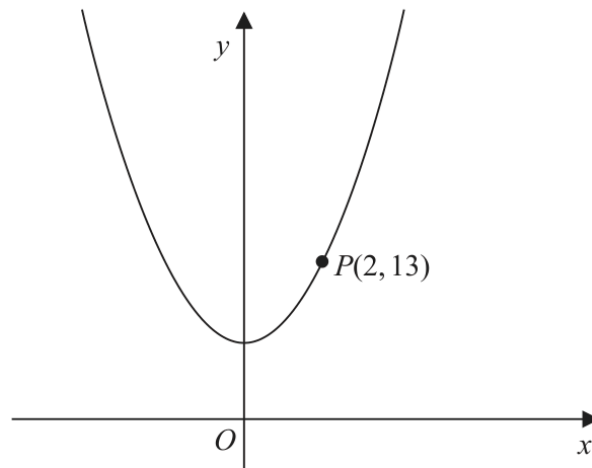


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2, 13)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at P .

(2)

The point Q with x coordinate $2 + h$ also lies on the curve.

(b) Find, in terms of h , the gradient of the line PQ . Give your answer in simplest form.

(3)

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)

(Total 6 marks)

Worked Solution - Question 7

1. Differentiate for the tangent gradient

For $y = 2x^2 + 5$, $\frac{dy}{dx} = 4x$. At P , $x = 2$, so the tangent gradient is 8.

2. Find the coordinates of Q

If $x_Q = 2 + h$, then $y_Q = 2(2 + h)^2 + 5 = 13 + 8h + 2h^2$.

3. Find the gradient of PQ

$$\text{gradient } PQ = \frac{(13 + 8h + 2h^2) - 13}{(2 + h) - 2} = \frac{8h + 2h^2}{h} = 8 + 2h.$$

4. Explain the link

As h gets closer to 0, point Q gets closer to P , and the chord gradient $8 + 2h$ tends to the tangent gradient 8.

Final answer

(a) 8.

(b) $8 + 2h$.

(c) As $h \rightarrow 0$, $8 + 2h \rightarrow 8$.

Question 8

Quadratics

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a , b and c are integers to be found.

(5)

(Total 5 marks)

Worked Solution - Question 8

1. Use a substitution

Let $u = \sqrt{x}$. Then $x = u^2$, so the equation becomes $u^2 - 6u + 4 = 0$.

2. Solve the quadratic

$$u = \frac{6 \pm \sqrt{36 - 16}}{2} = \frac{6 \pm 2\sqrt{5}}{2} = 3 \pm \sqrt{5}.$$

3. Return to x

Since $u = \sqrt{x}$, $x = u^2 = (3 \pm \sqrt{5})^2 = 9 + 5 \pm 6\sqrt{5} = 14 \pm 6\sqrt{5}$.

Final answer

$$x = 14 + 6\sqrt{5} \text{ or } x = 14 - 6\sqrt{5}.$$

Question 9

Trigonometric Functions

9.

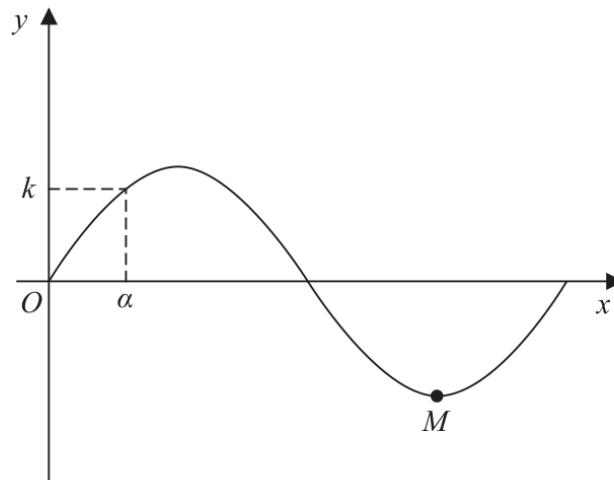


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C .

(a) State the period of C .

(1)

(b) State the coordinates of M .

(1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α .

Find, in terms of α ,

(c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,

(ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$.

(2)

(Total 4 marks)

Worked Solution - Question 9

1. Use the scale factor inside the sine

For $y = \sin\left(\frac{x}{12}\right)$, one full sine cycle needs $\frac{x}{12}$ to increase by 2π , so the period is 24π .

2. Locate the minimum point

The sine function is -1 when its angle is $\frac{3\pi}{2}$. Therefore $\frac{x}{12} = \frac{3\pi}{2}$, so $x = 18\pi$ and $M = (18\pi, -1)$.

3. Use sine symmetry

If the smallest positive solution of $\sin\left(\frac{x}{12}\right) = k$ is α , then the closest negative solution is one half-period before the opposite sine position: $-12\pi - \alpha$.

4. Convert to cosine

Since $\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$, the corresponding smallest positive cosine solution is $6\pi - \alpha$.

Final answer

(a) 24π .

(b) $(18\pi, -1)$.

(c)(i) $-12\pi - \alpha$. (ii) $6\pi - \alpha$.

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10 marks

Question 10

Graphs of Functions

10.

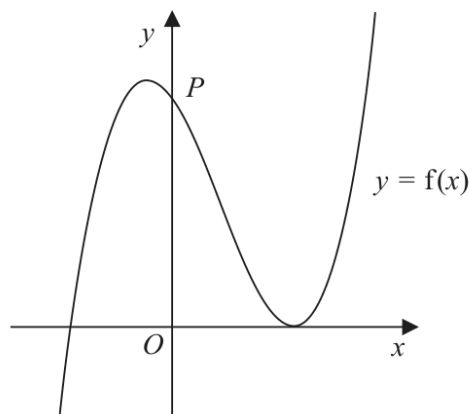


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

- (a) Deduce the values of x for which $f(x) \leq 0$ (2)

The curve crosses the y -axis at the point P , as shown.

- (b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

- (c) Hence, or otherwise, find

- (i) the coordinates of P ,
(ii) the gradient of the curve at P .

(2)

The curve with equation $y = f(x)$ is translated two units in the positive x direction to a curve with equation $y = g(x)$.

- (d) (i) Find $g(x)$, giving your answer in a simplified factorised form.

- (ii) Hence state the y intercept of the curve with equation $y = g(x)$.

(3)

(Total 10 marks)

Worked Solution - Question 10

1. Use the factorised form for the inequality

$f(x) = (2x + 5)(x - 3)^2$. Since $(x - 3)^2 \geq 0$, the sign is controlled by $2x + 5$, with an extra zero at $x = 3$. Hence $f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$.

2. Expand the cubic

$(x - 3)^2 = x^2 - 6x + 9$, so

$$f(x) = (2x + 5)(x^2 - 6x + 9) = 2x^3 - 7x^2 - 12x + 45.$$

3. Find P and the gradient at P

At the y-axis, $x = 0$, so $P = (0, 45)$. Also $f'(x) = 6x^2 - 14x - 12$, so the gradient at P is $f'(0) = -12$.

4. Translate the graph

A translation 2 units in the positive x direction means $g(x) = f(x - 2)$. Therefore $g(x) = (2(x - 2) + 5)((x - 2) - 3)^2 = (2x + 1)(x - 5)^2$.

5. Find the new y-intercept

Set $x = 0$: $g(0) = (1)(25) = 25$, so the y-intercept is 25.

Final answer

$f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$; $f(x) = 2x^3 - 7x^2 - 12x + 45$;
 $P = (0, 45)$; gradient = -12 ; $g(x) = (2x + 1)(x - 5)^2$; y-intercept 25.

Question 11

Integration

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P , writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) $f(x)$.

(8)

(Total 10 marks)

Worked Solution - Question 11

1. Use the given gradient for the tangent

At $P\left(4, \frac{32}{3}\right)$, $f'(4) = 5$, so the tangent has gradient 5.

2. Find the tangent equation

$y - \frac{32}{3} = 5(x - 4)$. Hence $y = 5x - \frac{28}{3}$.

3. Integrate f'' to find f'

$f''(x) = 4x^{-1/2} - 3$, so $f'(x) = 8x^{1/2} - 3x + C$.

4. Use $f'(4) = 5$

$5 = 8(2) - 12 + C = 4 + C$, so $C = 1$. Therefore $f'(x) = 8\sqrt{x} - 3x + 1$.

5. Integrate again

$f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x + D$.

6. Use the point on the curve

Substitute $x = 4$, $f(4) = \frac{32}{3}$: $\frac{32}{3} = \frac{16}{3}(8) - \frac{3}{2}(16) + 4 + D = \frac{68}{3} + D$, so $D = -12$.

Final answer

(a) $y = 5x - \frac{28}{3}$.

(b) $f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x - 12$.